• Transformarea Laplace

1. Determinați transformările Laplace ale următoarelor funcții original:

(a) $(t^2-1)e^{2t}$
(b) $t\cos t e^{-t}$
(c) $\operatorname{ch} t \cos t$
(d) $t\sin^2 t$
(e) $e^{-t}\cos 2t$
(f) $(t^2-t+1)\operatorname{ch} t$.

Rezolvare

(a) $\mathcal{L}((t^2-1)e^{2t})(s) = \mathcal{L}(t^2e^{2t})(s) - \mathcal{L}(e^{2t})(s)$

$\mathcal{L}(t^2e^{2t})(s) = \mathcal{L}(t^2)(s-2) = \frac{2!}{(s-2)^3} = \frac{2}{(s-2)^3}$

$\mathcal{L}(e^{2t})(s) = \frac{1}{s-2}$

$\Rightarrow \mathcal{L}(f)(s) = \frac{2}{(s-2)^3} - \frac{1}{s-2} = \frac{2-(s-2)^2}{(s-2)^3} = \frac{-s^2+4s-2}{(s-2)^3}$

(b) $\mathcal{L}(t\cos t e^{-t})(s) = -\frac{d}{ds}\mathcal{L}(\cos t e^{-t})(s)$

$\mathcal{L}(\cos t e^{-t})(s) = \mathcal{L}(\cos t)(s+1) = \frac{s+1}{(s+1)^2+1}$

$\Rightarrow \mathcal{L}(f)(s) = -\frac{d}{ds}\frac{s+1}{(s+1)^2+1} =$

$= -\frac{(s+1)^2+1-(s+1)\cdot 2(s+1)}{[(s+1)^2+1]^2} = -\frac{1-(s+1)^2}{(s^2+2s+2)^2} = \frac{s^2+2s}{(s^2+2s+2)^2}$

(c) $\mathcal{L}(\operatorname{ch} t \cos t)(s) = \mathcal{L}\left(\frac{e^t+e^{-t}}{2}\cos t\right)(s) =$

$= \frac{1}{2}\left(\mathcal{L}(e^t\cos t)(s) + \mathcal{L}(e^{-t}\cos t)(s)\right)$

$\mathcal{L}(e^t\cos t)(s) = \mathcal{L}(\cos t)(s-1) = \frac{s-1}{(s-1)^2+1}$

$\mathcal{L}(e^{-t}\cos t)(s) = \mathcal{L}(\cos t)(s+1) = \frac{s+1}{(s+1)^2+1}$

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$$\Rightarrow \mathcal{L}(f)(s) = \frac{1}{2}\left[\frac{s-1}{(s-1)^2+1} + \frac{s+1}{(s+1)^2+1}\right] = \frac{1}{2}\left(\frac{s-1}{s^2-2s+2} + \frac{s+1}{s^2+2s+2}\right)$$

$$= \frac{1}{2}\cdot\frac{(s-1)(s^2+2s+2) + (s+1)(s^2-2s+2)}{(s^2-2s+2)(s^2+2s+2)} =$$

$$= \frac{1}{2}\cdot\frac{s^3-s^2+2s^2-2s+2s-2+s^3+s^2-2s^2-2s+2s+2}{s^4-2s^3+2s^2+2s^3-4s^2-4s+2s^2+4s+4} =$$

$$= \frac{1}{2}\cdot\frac{2s^3}{s^4+4} = \frac{s^3}{s^4+4}$$

(d) $\mathcal{L}(t\sin^2 t)(s) = -\frac{d}{ds}\mathcal{L}(\sin^2 t)(s)$

$$\mathcal{L}(\sin^2 t)(s) = \mathcal{L}\left(\frac{1-\cos 2t}{2}\right)(s) = \frac{1}{2}\mathcal{L}(H)(s) - \frac{1}{2}\mathcal{L}(\cos 2t)(s)$$

$$= \frac{1}{2s} - \frac{1}{2}\cdot\frac{s}{s^2+2} = \frac{1}{2}\left(\frac{1}{s} - \frac{s}{s^2+2}\right) = \frac{1}{2}\cdot\frac{s^2+2-s^2}{s(s^2+2)} =$$

$$= \frac{1}{s(s^2+2)}$$

$$\Rightarrow \mathcal{L}(f)(s) = -\frac{d}{ds}\cdot\frac{1}{s(s^2+2)} = +\frac{3s^2+2}{s^2(s^2+2)^2}$$

(e) $\mathcal{L}(e^{-t}\cos 2t)(s) = \mathcal{L}(\cos 2t)(s+1) = \frac{s+1}{(s+1)^2+4} =$

$$= \frac{s+1}{s^2+2s+5}$$

(f) $\mathcal{L}((t^2-t+1)\operatorname{ch} t)(s) = \mathcal{L}(t^2\operatorname{ch} t)(s) - \mathcal{L}(t\operatorname{ch} t)(s) + \mathcal{L}(\operatorname{ch} t)(s)$

$$\mathcal{L}(t^2\operatorname{ch} t)(s) = \mathcal{L}\left(t^2\cdot\frac{e^t+e^{-t}}{2}\right)(s) = \frac{1}{2}\mathcal{L}(e^t\cdot t^2)(s) + \frac{1}{2}\mathcal{L}(e^{-t}\cdot t^2)(s) =$$

$$= \frac{1}{(s-1)^3} + \frac{1}{(s+1)^3}$$

$$\mathcal{L}(e^t\cdot t^2)(s) = \mathcal{L}(t^2)(s-1) = \frac{2!}{(s-1)^3}$$

$$\mathcal{L}(e^{-t}\cdot t^2)(s) = \mathcal{L}(t^2)(s+1) = \frac{2!}{(s+1)^3}$$

$$\mathcal{L}(t\operatorname{ch} t)(s) = \mathcal{L}\left(t\cdot\frac{e^t+e^{-t}}{2}\right)(s) = \frac{1}{2}\mathcal{L}(te^t)(s) + \frac{1}{2}\mathcal{L}(te^{-t})(s)$$

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$$\mathcal{L}(t\cdot e^t)(s) = \mathcal{L}(t)(s-1) = \frac{1}{(s-1)^2}$$

$$\mathcal{L}(t\cdot e^{-t})(s) = \mathcal{L}(t)(s+1) = \frac{1}{(s+1)^2}$$

$$\Rightarrow \mathcal{L}(t\operatorname{ch} t)(s) = \left[\frac{1}{(s-1)^2} + \frac{1}{(s+1)^2}\right]\cdot\frac{1}{2}$$

$$\mathcal{L}(\operatorname{ch} t)(s) = \mathcal{L}\left(\frac{e^t+e^{-t}}{2}\right)(s) = \frac{1}{2}\mathcal{L}(e^t)(s) + \frac{1}{2}\mathcal{L}(e^{-t})(s) =$$

$$= \frac{1}{2}\left(\frac{1}{s-1} + \frac{1}{s+1}\right)$$

$$\Rightarrow \mathcal{L}(f)(s) = \frac{1}{(s-1)^3} + \frac{1}{(s+1)^3} - \frac{1}{2}\left[\frac{1}{(s-1)^2} + \frac{1}{(s+1)^2}\right] + \frac{1}{2}\left(\frac{1}{s-1} + \frac{1}{s+1}\right)$$

$$= \frac{1}{(s-1)^3} + \frac{1}{(s+1)^3} - \frac{1}{2}\,\frac{s^2+2s+1+s^2-2s+1}{(s^2-1)^2} + \frac{1}{2}\cdot\frac{s+1+s-1}{s^2-1} =$$

$$= \frac{(s+1)^3+(s-1)^3}{(s^2-1)^3} - \frac{s^2+1}{(s^2-1)^2} + \frac{s}{s^2-1} =$$

$$= \frac{(s+1+s-1)\left((s+1)^2-(s+1)(s-1)+(s-1)^2\right) - (s^2+1)(s^2-1) + s(s^2-1)^2}{(s^2-1)^3} =$$

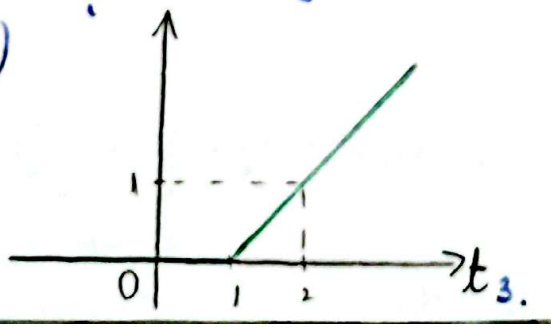
$$= \frac{2s\left(s^2+2s+1-s^2+1+s^2-2s+1\right) - s^4+1+s(s^4-2s^2+1)}{(s^2-1)^3} =$$

$$= \frac{2s(s^2+3) - s^4+1+s^5-2s^3+s}{(s^2-1)^3} = \frac{2s^3+6s-s^4+1+s^5-2s^3+s}{(s^2-1)^3} =$$

$$= \frac{s^5-s^4+7s+1}{(s^2-1)^3}$$

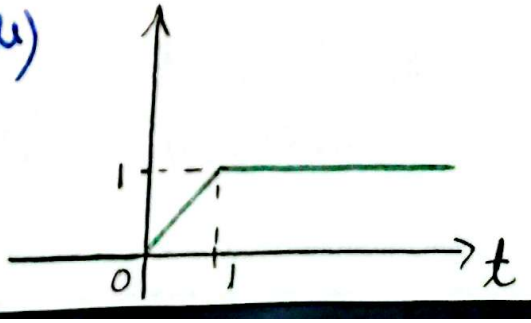
(2.) Exprimați următoarele funcții original având graficele indicate cu ajutorul funcției Heaviside și determinați transformatele Laplace ale acestora.

(a)



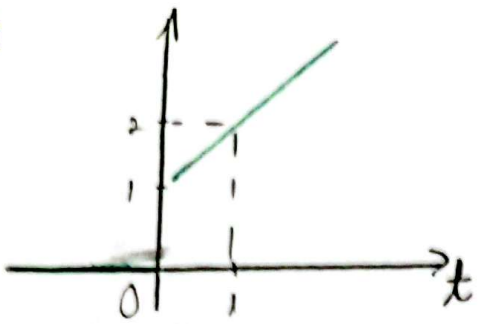
3.

(b)

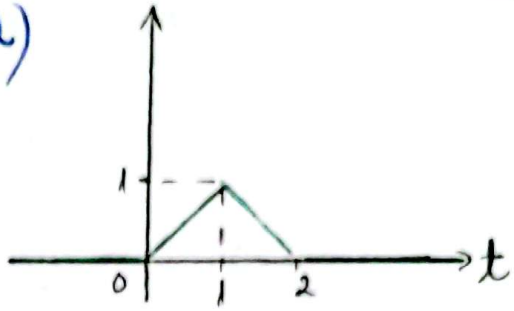


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(c)



(d)



Rezolvare

(a) $f(t) = (t-1)H(t-1)$

$\mathcal{L}(f)(s) = \mathcal{L}\left((t-1)H(t-1)\right)(s) = \mathcal{L}\left(tH(t-1)\right)(s) - \mathcal{L}(H(t-1))(s)$

$\mathcal{L}\left(tH(t-1)\right)(s) = -\frac{d}{ds}\mathcal{L}\left(H(t-1)\right)(s) = -\frac{d}{ds}\cdot\frac{e^{-s}}{s} =$

$= -\frac{-se^{-s}-e^{-s}}{s^2} = \frac{e^{-s}(s+1)}{s^2}$

$\mathcal{L}\left(H(t-1)\right)(s) = \frac{e^{-s}}{s}$

$\Rightarrow \mathcal{L}(f)(s) = \frac{e^{-s}(s+1)}{s^2} - \frac{e^{-s}}{s} = \frac{e^{-s}(s+1-s)}{s^2} = \frac{e^{-s}}{s^2}$

(b) $f(t) = t(H(t) - H(t-1)) + H(t-1)$

$\mathcal{L}(f)(s) = \mathcal{L}\left[t(H(t) - H(t-1)) + H(t-1)\right](s) =$

$= \mathcal{L}\left(tH(t)\right)(s) - \mathcal{L}\left(tH(t-1)\right)(s) + \mathcal{L}\left(H(t-1)\right)(s)$

$\mathcal{L}\left(tH(t)\right)(s) = -\frac{d}{ds}\mathcal{L}(H(t))(s) = -\frac{d}{ds}\cdot\frac{1}{s} = \frac{1}{s^2}$

$\mathcal{L}\left(tH(t-1)\right)(s) = -\frac{d}{ds}\mathcal{L}\left(H(t-1)\right)(s) = -\frac{d}{ds}\frac{e^{-s}}{s} = \frac{e^{-s}(s+1)}{s^2}$

$\mathcal{L}\left(H(t-1)\right)(s) = \frac{e^{-s}}{s}$

$\Rightarrow \mathcal{L}(f)(s) = \frac{1}{s^2} - \frac{e^{-s}(s+1)}{s^2} + \frac{e^{-s}}{s} = \frac{1-e^{-s}}{s^2}$

(c) $f(t) = t+1$

$\mathcal{L}(t+1)(s) = \mathcal{L}(t)(s) + \mathcal{L}(1)(s) = \frac{1}{s^2} + \frac{1}{s} = \frac{s+1}{s^2}$

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(d) $f(t) = t(H(t) - H(t-1)) + (2-t)(H(t-1) - H(t-2))$

$\mathcal{L}(f)(s) = \mathcal{L}[t(H(t) - H(t-1)) + (2-t)(H(t-1) - H(t-2))](s) =$

$= \mathcal{L}(tH(t))(s) - \mathcal{L}(tH(t-1))(s) + 2\mathcal{L}(H(t-1))(s) - \mathcal{L}(tH(t-1))(s) - 2\mathcal{L}(H(t-2))(s)$

$+ \mathcal{L}(tH(t-2))(s)$

$\mathcal{L}(tH(t))(s) = -\frac{d}{ds}\mathcal{L}(H(t))(s) = -\frac{d}{ds}\cdot\frac{1}{s} = \frac{1}{s^2}$

$\mathcal{L}(tH(t-1))(s) = -\frac{d}{ds}\mathcal{L}(H(t-1))(s) = -\frac{d}{ds}\frac{e^{-s}}{s} = \frac{e^{-s}(s+1)}{s}$

$\mathcal{L}(H(t-1))(s) = \frac{e^{-s}}{s}$

$\mathcal{L}(H(t-2))(s) = \frac{e^{-2s}}{s}$

$\mathcal{L}(tH(t-2))(s) = -\frac{d}{ds}\mathcal{L}(H(t-2))(s) = -\frac{d}{ds}\cdot\frac{e^{-2s}}{s} =$

$= -\frac{d}{ds}\frac{-2e^{-2s}\cdot s - e^{-2s}}{s^2} = +\frac{d}{ds}\frac{e^{-2s}(2s+1)}{s^2}$

$\Rightarrow \mathcal{L}(f)(s) = \frac{1}{s^2} - \frac{2e^{-s}(s+1)}{s} + \frac{2e^{-s}}{s} - \frac{2e^{-2s}}{s} + \frac{e^{-2s}(2s+1)}{s^2} =$

$= \frac{1 - 2se^{-s}(s+1) + 2se^{-s} - 2se^{-2s} + e^{-2s}(2s+1)}{s^2} =$

$= \frac{1 + e^{-s}(2s - 2s^2 - 2s) + e^{-2s}(2s + 1 - 2s)}{s^2} =$

$= \frac{e^{-2s} - 2s^2e^{-s} + 1}{s^2}$

(3.) Determinați funcțiile original ale căror transformate Laplace sunt:

(a) $F(s) = \frac{1}{s(s+1)(s+2)}$

(b) $F(s) = \frac{1}{(s^2-2)^2}$

(c) $F(s) = \frac{2s}{s^4+1}$

(d) $F(s) = \frac{6}{s^4+10s^2+9}$

(e) $F(s) = \frac{5}{(s+2)^4}$

(f) $F(s) = \frac{3s+2}{s^2+2s+10}$

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Rezolvare

(a) $F(s) = \mathcal{L}(f)(s) = \frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} =$

$= \frac{A(s+1)(s+2) + B \cdot s(s+2) + C \cdot s(s+1)}{s(s+1)(s+2)} \Rightarrow$

$\Rightarrow 1 = s^2(A+B+C) + s(3A+2B+C) + 2A \Rightarrow$

$\Leftrightarrow \begin{cases} 2A = 1 \Rightarrow A = \frac{1}{2} \\ 3A + 2B + C = 0 \\ A + B + C = 0 \end{cases} \Rightarrow \begin{cases} 2B + C = -\frac{3}{2} \\ B + C = -\frac{1}{2} \end{cases} \Rightarrow B = -1 \Rightarrow = \frac{1}{2}$

$\Rightarrow F(s) = \mathcal{L}(f)(s) = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{s+2} = \mathcal{L}\left(\frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}\right)(s)$

$\Rightarrow f(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$

(b) $F(s) = \mathcal{L}(f)(s) = \frac{1}{(s^2-1)^2} = \frac{1}{(s-1)^2(s+1)^2} \Rightarrow s_0 = 1$ pol dublu, $s_1 = -1$ pol dublu

$\Rightarrow f(s) = \text{Rez}\left(F(s) \cdot e^{st}, s_0\right) + \text{Rez}\left(F(s) \cdot e^{st}, s_1\right)$

$\text{Rez}\left(F(s) \cdot e^{st}, s_0\right) = \lim_{s \to 1} \left[\cancel{(s-1)^2} \cdot \frac{e^{st}}{\cancel{(s-1)^2}(s+1)^2} \right]' =$

$= \lim_{s \to 1} \left[\frac{e^{st}}{(s+1)^2} \right]' = \lim_{s \to 1} \frac{t \cdot e^{st}(s+1)^2 - e^{st} \cdot 2 \cdot (s+1)}{(s+1)^4} =$

$= \frac{t \cdot e^t \cdot 4 - e^t \cdot 4}{16} = \frac{1}{4} e^t (t-1)$

$\text{Rez}\left(F(s) \cdot e^{st}, s_1\right) = \lim_{s \to -1} \left[\cancel{(s+1)^2} \cdot \frac{e^{st}}{(s-1)^2 \cancel{(s+1)^2}} \right]' =$

$= \lim_{s \to -1} \left[\frac{e^{st}}{(s-1)^2} \right]' = \lim_{s \to -1} \frac{t e^{st}(s-1)^2 - e^{st} \cdot 2(s-1)}{(s-1)^4} =$

$= \frac{t \cdot e^{-t} \cdot 4 + e^{-t} \cdot 4}{16} = \frac{1}{4} e^{-t}(t+1)$

$\Rightarrow f(t) = \frac{1}{4}\left[e^t(t-1) + e^{-t}(t+1)\right] = \frac{1}{4}\left(te^t - e^t + te^{-t} + e^{-t}\right) =$

$= \frac{1}{2}\left(t \frac{e^t + e^{-t}}{2} - \frac{e^t - e^{-t}}{2}\right) = \frac{1}{2}(t \operatorname{ch} t - \operatorname{sh} t)$

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(c) $F(s) = \mathcal{L}(f)(s) = \frac{2s}{s^4+1} = \frac{2s}{s^4+2s^2+1-2s^2} = \frac{2s}{(s^2+1)^2-2s^2} =$

$= \frac{2s}{(s^2+s\sqrt{2}+1)(s^2-s\sqrt{2}+1)} = \frac{As+B}{s^2+s\sqrt{2}+1} + \frac{Cs+D}{s^2-s\sqrt{2}+1} =$

$= \frac{(As+B)(s^2-s\sqrt{2}+1)+(Cs+D)(s^2+s\sqrt{2}+1)}{(s^2+s\sqrt{2}+1)(s^2-s\sqrt{2}+1)}$

$\Rightarrow 2s = As^3+Bs^2-s^2A\sqrt{2}-sB\sqrt{2}+As+B+Cs^3+Ds^2+sC\sqrt{2}+sD\sqrt{2}+$

$+Cs+D$

$= s^3(A+C)+s^2(B-A\sqrt{2}+D+C\sqrt{2})+s(A-B\sqrt{2}+D\sqrt{2}+C)+B+D$

$\Rightarrow \begin{cases} A+C=0 \Rightarrow A=-C \\ B-A\sqrt{2}+D+C\sqrt{2}=0 \Rightarrow 2C\sqrt{2}=0 \Rightarrow C=0 \Rightarrow A=0 \\ A-B\sqrt{2}+C+D\sqrt{2}=2 \\ B+D=0 \Rightarrow B=-D \end{cases} \Rightarrow 2D\sqrt{2}=2 \Rightarrow D=\frac{1}{\sqrt{2}} \Rightarrow B=-\frac{1}{\sqrt{2}}$

$\Rightarrow F(s) = \frac{-\frac{1}{\sqrt{2}}}{s^2+s\sqrt{2}+1} + \frac{\frac{1}{\sqrt{2}}}{s^2-s\sqrt{2}+1}$

$s^2+s\sqrt{2}+1 = s^2+2\cdot\frac{1}{\sqrt{2}}\cdot s+\left(\frac{1}{\sqrt{2}}\right)^2-\frac{1}{2}+1 = \left(s+\frac{1}{\sqrt{2}}\right)^2+\frac{1}{2}$

$s^2-s\sqrt{2}+1 = s^2-2\cdot\frac{1}{\sqrt{2}}\cdot s+\left(\frac{1}{\sqrt{2}}\right)^2-\frac{1}{2}+1 = \left(s-\frac{1}{\sqrt{2}}\right)^2+\frac{1}{2}$

$\Rightarrow F(s) = -\frac{1}{\sqrt{2}}\cdot\frac{1}{\left(s+\frac{1}{\sqrt{2}}\right)^2+\frac{1}{2}} + \frac{1}{\sqrt{2}}\cdot\frac{1}{\left(s-\frac{1}{\sqrt{2}}\right)^2+\frac{1}{2}} =$

$= -\mathcal{L}\left(\sin\frac{t}{\sqrt{2}}\right)(s+\frac{1}{\sqrt{2}}) + \mathcal{L}\left(\sin\frac{t}{\sqrt{2}}\right)(s-\frac{1}{\sqrt{2}}) =$

$= -e^{s/\sqrt{2}}\mathcal{L}\left(\sin\frac{t}{\sqrt{2}}\right)(s) + e^{-s/\sqrt{2}}\mathcal{L}\left(\sin\frac{t}{\sqrt{2}}\right)(s) = \dots$

(d) $F(s) = \frac{6}{s^4+10s^2+9}$

$s^4+10s^2+9=0 \xrightarrow{s^2=u} u^2+10u+9=0 \Rightarrow (u+1)(u+9)=0 \Rightarrow$

$\Rightarrow u_1=-1 \Rightarrow s^2=-1 \Rightarrow s_{1,2}=\pm i$

$u_2=-9 \Rightarrow s^2=-9 \Rightarrow s_{3,4}=\pm 3i$

$\Rightarrow F(s) = \sum_{k=1}^{4} \operatorname{Rez}(f, s_k) \; ; \; f(s) = F(s)\cdot e^{st}.$

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$$\text{Rez}(f, s_1) = \lim_{s\to i}(s-i)\cdot\frac{6e^{st}}{s^4+10s^2+9} = \lim_{s\to i}\frac{6e^{st}}{(s+i)(s+3i)(s-3i)}$$

$$= \frac{6e^{it}}{2i\cdot 4i(-2i)} = \frac{3}{8}\cdot\frac{e^{it}}{i}$$

$$\text{Rez}(f, s_2) = \lim_{s\to -i}(s+i)\frac{6e^{st}}{s^4+10s^2+9} = \lim_{s\to -i}\frac{6e^{st}}{(s-i)(s+3i)(s-3i)} =$$

$$= \frac{6e^{-it}}{(-2i)2i(-4i)} = \frac{-3}{8}\frac{e^{-it}}{i}$$

$$\text{Rez}(f, s_3) = \lim_{s\to 3i}(s-3i)\cdot\frac{6e^{st}}{s^4+10s^2+9} = \lim_{s\to 3i}\frac{6\cdot e^{st}}{(s-i)(s+i)(s+3i)} =$$

$$= \frac{6e^{3it}}{2i\cdot 4i\cdot 6i} = \frac{-1}{8}\cdot\frac{e^{3it}}{i}$$

$$\text{Rez}(f, s_4) = \lim_{s\to -3i}(s+3i)\frac{6e^{st}}{s^4+10s^2+9} = \lim_{s\to -3i}\frac{6e^{st}}{(s-i)(s+i)(s-3i)} =$$

$$= \frac{6e^{-3it}}{(-4i)(-2i)(-6i)} = \frac{1}{8}\cdot\frac{e^{-3it}}{i}$$

$$\Rightarrow f(t) = \frac{3}{8i}\left(e^{it} - e^{-it}\right) + \frac{1}{8i}\left(e^{-3it} - e^{3it}\right) =$$

$$= \frac{3}{8i}\left(\cancel{\cos t} + i\sin t - \cancel{\cos t} + i\sin t\right) + \frac{1}{8i}\left(\cos 3t - i\sin 3t - \right.$$

$$\left. - \cancel{\cos 3t} - i\sin 3t\right) =$$

$$= \frac{3}{8i}\cdot 2i\sin t + \frac{-1}{8i}\,2i\sin 3t =$$

$$= \frac{3}{4}\sin t - \frac{1}{4}\sin 3t.$$

(e) $F(s) = \dfrac{5}{(s+2)^4}$; $s = -2$ pol de ordin 4

$$\Rightarrow f(t) = \text{Rez}\left(F(s)\cdot e^{st}; s\right) = \frac{1}{3!}\lim_{s\to -2}\left[\cancel{(s+2)^4}\cdot\frac{e^{st}\cdot 5}{\cancel{(s+2)^4}}\right]''' =$$

$$= \frac{1}{6}\lim_{s\to -2}(5e^{st})''' = \frac{1}{6}\lim_{s\to -2} 5\cdot t^3 e^{st} = \frac{5}{6}t^3\cdot e^{-2t}$$

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(4) Utilizând transformarea Laplace, determinați soluțiile următoarelor ecuații și sisteme de ecuații diferențiale în mulțimea funcțiilor original:

(a) $x'' + 6x' + 13x = 0$; $x(0_+) = x'(0_+) = 1$

(b) $x'' + 4x' + 4x = te^{-2t}$; $x(0_+) = 1$; $x'(0_+) = 2$

(c) $x'' - 2x' + 10x = e^t \cos 3t$; $x(0_+) = x'(0_+) = 1$

(d) $x'' + x = H(t) - H(t - 2\pi)$; $x(0_+) = 1$; $x'(0_+) = 0$

(e) $\begin{cases} x' + 4x - y = -36t \\ y' + 2x - y = -2e^t \end{cases}$; $x(0_+) = 0$; $y(0_+) = 1$

(f) $x(t) = t + \int_0^t \sin(t - \tau) x(\tau)\, d\tau$.

Rezolvare $\left[\mathcal{L}(f^{(n)})(s) = s^n \mathcal{L}(f)(s) - \left(s^{n-1} f(0_+) + \dots + s f^{(n-2)}(0_+) + f^{(n-1)}(0_+)\right)\right]$

(a) Fie $F(s)$ not $\mathcal{L}[x(t)](s)$. Aplicând tr. Laplace ec $x'' + 6x' + 13x = 0$ obținem:

$$\mathcal{L}[x''(t)](s) + 6\mathcal{L}[x'(t)](s) + 13\mathcal{L}[x(t)](s) = 0$$

$$s^2 F(s) - sx(0_+) - x'(0_+) + 6\left(sF(s) - x(0_+)\right) + 13F(s) = 0$$

$$F(s)(s^2 + 6s + 13) = s + 7.$$

$$F(s) = \frac{s+7}{s^2+6s+13}$$

$s^2 + 6s + 13 = 0$; $\Delta = 36 - 52 = -16 \Rightarrow \sqrt{\Delta} = 4i$

$$s_{1,2} = \frac{-6 \pm 4i}{2} = -3 \pm 2i$$

$$\Rightarrow x(t) = \operatorname{Rez}\left(F(s) \cdot e^{st}; -3+2i\right) + \operatorname{Rez}\left(F(s) \cdot e^{st}; -3-2i\right)$$

$$\operatorname{Rez}\left(F(s) \cdot e^{st}; -3+2i\right) = \lim_{s \to -3+2i} (s+3-2i) \cdot \frac{(s+7) \cdot e^{st}}{s^2+6s+13} =$$

$$= \lim_{s \to -3+2i} \frac{(s+7)e^{st}}{s+3+2i} = \frac{(4+2i) \cdot e^{-3t} \cdot e^{2it}}{4i}$$

$$\operatorname{Rez}\left(F(s) \cdot e^{st}; -3-2i\right) = \lim_{s \to -3-2i} (s+3+2i) \cdot \frac{(s+7)e^{st}}{s^2+6s+13} =$$

$$= \lim_{s \to -3-2i} \frac{(s+7) \cdot e^{st}}{s+3-2i} = \frac{(4-2i)e^{-3t} \cdot e^{-2it}}{-4i}$$

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$$\Rightarrow x(t) = \frac{(4+2i)}{4i} e^{3t} \cdot e^{2it} - \frac{(4-2i)e^{-3t} \cdot e^{-2it}}{4i} =$$

$$= \frac{2e^{-3t}}{4i}\left[(2+i)(\cos 2t + i\sin 2t) - (2-i)(\cos 2t - i\sin 2t)\right] =$$

$$= \frac{e^{-3t}}{2i}\left(2\cos 2t + i\cos 2t + 2i\sin 2t - \sin 2t - 2\cos 2t + i\cos 2t + \right.$$

$$\left. + 2i\sin 2t + \sin 2t\right) =$$

$$= e^{-3t}\left(2\sin 2t + \cos 2t\right)$$

(b) Fie $F(s) \overset{not}{=} \mathcal{L}[x(t)](s)$. Aplicând tr. Laplace ec $x'' + 4x' + 4x = te^{-2t}$ obținem:

$$\mathcal{L}[x''(t)](s) + 4\mathcal{L}[x'(t)](s) + 4\mathcal{L}[x(t)](s) = \mathcal{L}[te^{-2t}](s)$$

$$\mathcal{L}[te^{-2t}](s) = \mathcal{L}(t)(s+2) = \frac{1}{(s+2)^2}$$

$$\Rightarrow s^2F(s) - s\cdot x(0_+) - x'(0_+) + 4\left(sF(s) - x(0_+)\right) + 4F(s) = \frac{1}{(s+2)^2}$$

$$F(s)(s^2+4s+4) + s + 2 + 4 = \frac{1}{(s+2)^2}$$

$$F(s)(s+2)^2 = \frac{1}{(s+2)^2} + s + 6 = \frac{1 + (s+6)(s+2)^2}{(s+2)^2}$$

$$F(s) = \frac{1 + (s+6)(s^2+4s+4)}{(s+2)^4} = \frac{1 + (s^3+6s^2+4s^2+24s+4s+24)}{(s+2)^4} =$$

$$= \frac{s^3+10s^2+28s+25}{(s+2)^4}$$

$$\Rightarrow x(t) = \operatorname{Rez}\left(F(s)\cdot e^{st}; -2\right) = \lim_{s\to -2} \frac{1}{3!}\left[(s+2)^4 \frac{(s^3+10s^2+28s+25)e^{st}}{(s+2)^4}\right]''$$

$$= \frac{1}{6}\lim_{x\to -2}\left[(s^3+10s^2+28s+25)e^{st}\right]'' =$$

$$= \frac{1}{6}\lim_{x\to -2}\left[(3s^2+20s+28)e^{st} + te^{st}(s^3+10s^2+28s+25)\right]' =$$

$$= \frac{1}{6}\lim_{x\to -2}\left[e^{st}(s^3+13s^2+48s+53)\right]'' =$$

$$= \frac{1}{6}\lim_{x\to -2}\left[e^{st}(s^3+13s^2+48s+53+3s^2+26s+48)\right]' =$$

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$$= \frac{1}{6} \lim_{x \to -2} \left[e^{st} \left(ts^3 + (3+10t)s^2 + (20+28t)s + 28 + 25t \right) \right]'' =$$

$$= \frac{1}{6} \lim_{x \to -2} \left[e^{st} \left(3ts^2 + (6+20t)s + 20 + 28t \right) + te^{st} \left(ts^3 + (3+10t)s^2 + \right. \right.$$

$$\left. \left. + (20+28t)s + 28 + 25t \right] \right]' =$$

$$= \frac{1}{6} \lim_{x \to -2} \left[e^{st} \left(t^2 s^3 + (6t+10t^2)s^2 + (6+40t+28t^2)s + 20+56t+25t^2 \right) \right]$$

$$= \frac{1}{6} \lim_{x \to -2} \left[e^{st} \left(3t^2 s^2 + (12t+20t^2)s + 6+40t+28t^2 \right) + te^{st} \left(t^2 s^3 + \right. \right.$$

$$\left. \left. + (6t+10t^2)s^2 + (6+40t+28t^2)s + 20+56t+25t^2 \right) \right] =$$

$$= \frac{1}{6} \lim_{x \to -2} e^{st} \left[s^3 \cdot t^3 + s^2 \left(\cancel{12t} + \cancel{20}^{9} t^2 + 10t^3 \right) + s \left(18t + 60t^2 + 28t^3 \right) + \right.$$

$$\left. + 6 + 60t + 84t^2 + 25t^3 \right] =$$

$$= \frac{1}{6} e^{-2t} \left[-8t^3 + 4(9t^2 + 10t^3) - 2(18t + 60t^2 + 28t^3) + 6 + 60t + 84t^2 + \right.$$

$$\left. + 25t^3 \right] =$$

$$= \frac{1}{6} e^{-2t} \left(-8t^3 + \cancel{36t^2} + 40t^3 - 36t - \cancel{120t^2} - 56t^3 + 6 + 60t + \cancel{84t^2} + 25t^3 \right) =$$

$$= \frac{1}{6} e^{-2t} \left(t^3 + 24t + 6 \right) = \frac{1}{6} t^3 e^{-2t} + 4te^{-2t} + e^{-2t}.$$

(c) Fie $F(s) \overset{not}{=} \mathcal{L}[x(t)](s)$. Aplicând tr Laplace ec.

$x'' - 2x' + 10x = e^t \cos 3t$ obținem:

$$\mathcal{L}[x''(t)](s) - 2\mathcal{L}[x'(t)](s) + 10\mathcal{L}[x(t)](s) = \mathcal{L}[e^t \cos 3t](s)$$

$$\mathcal{L}[e^t \cos 3t](s) = \mathcal{L}[\cos 3t](s-1) = \frac{s-1}{(s-1)^2 + 9}$$

$$s^2 F(s) - sx(0_+) - x'(0_+) - 2\left(sF(s) - x(0_+) \right) + 10F(s) = \frac{s-1}{(s-1)^2+9}$$

$$F(s)(s^2 - 2s + 10) - s - 1 + 2 = \frac{s-1}{s^2 - 2s + 10}$$

$$F(s) = \frac{s-1 + (s-1)(s^2-2s+10)}{(s^2-2s+10)^2} = \frac{(s-1)(s^2-2s+11)}{(s^2-2s+10)^2} =$$

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$$= \frac{s^3 - s^2 - 2s^2 + 2s + 11s - 11}{(s^2 - 2s + 10)^2} = \frac{s^3 - 3s^2 + 13s - 11}{(s^2 - 2s + 10)^2}$$

$s^2 - 2s + 10 = 0 \Rightarrow \Delta = 4 - 40 = -36 \Rightarrow \sqrt{\Delta} = 6i$

$s_{1,2} = \frac{2 \pm 6i}{2} = 1 \pm 3i$

$\Rightarrow y(t) = \sum_{k=1}^{2} \operatorname{Rez}\left(F(s) \cdot e^{st}; s_k\right)$

$\operatorname{Rez}\left(F(s) \cdot e^{st}; 1+3i\right) = \lim_{s \to 1+3i} \left[(s-1-3i)^2 \cdot \frac{(s^3 - 3s^2 + 13s - 11)e^{st}}{(s-1-3i)^2 (s-1+3i)^2} \right]^{(1)}$

$= \lim_{s \to 1+3i} \left[\frac{(s^3 - 3s^2 + 13s - 11)e^{st}}{(s-1+3i)^2} \right]^{(1)} = \dots\dots$

(a) Fie $F(s)$ not $\mathcal{L}[x(t)](s)$. Aplicând tr Laplace ec

$x'' + x = H(t) - H(t - 2\pi)$ obținem:

$\mathcal{L}[x''(t)](s) + \mathcal{L}[x(t)](s) = \mathcal{L}[H(t) - H(t-2\pi)](s)$

$\mathcal{L}[H(t) - H(t-2\pi)](s) = \mathcal{L}[H(t)](s) - \mathcal{L}[H(t-2\pi)](s) =$

$= \frac{1}{s} - \frac{e^{-2\pi s}}{s}$

$\Rightarrow s^2 F(s) - sx(0_+) - \cancel{x'(0_+)} + F(s) = \frac{1 - e^{-2\pi s}}{s}$

$F(s)(s^2+1) - s = \frac{1 - e^{-2\pi s}}{s}$

$F(s) = \frac{1 - e^{-2\pi s} + s^2}{s(s^2+1)} = \frac{s}{s^2+1} + \frac{1}{s^2+1} \cdot \frac{1 - e^{-2\pi s}}{s} \Rightarrow$

$\Rightarrow \mathcal{L}[x(t)](s) = \mathcal{L}[\cos t](s) + \mathcal{L}[\sin t](s) \cdot \mathcal{L}[H(t-2\pi)](s)$

$= \mathcal{L}[\cos t](s) + \mathcal{L}[\sin t * H(t-2\pi)](s) =$

$= [\dots] = \begin{cases} 1 & ; t < 2\pi \\ \cos t & ; t \geq 2\pi \end{cases}$.

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$$= \frac{s^3 - s^2 - 2s^2 + 2s + 11s - 11}{(s^2 - 2s + 10)^2} = \frac{s^3 - 3s^2 + 13s - 11}{(s^2 - 2s + 10)^2}$$

$s^2 - 2s + 10 = 0 \Rightarrow \Delta = 4 - 40 = -36 \Rightarrow \sqrt{\Delta} = 6i$

$s_{1,2} = \frac{2 \pm 6i}{2} = 1 \pm 3i$

$$\Rightarrow x(t) = \sum_{k=1}^{2} \operatorname{Rez}\left(F(s) \cdot e^{st}; s_k\right)$$

$$\operatorname{Rez}\left(F(s) \cdot e^{st}; 1+3i\right) = \lim_{x \to 1+3i} \left[(s-1-3i)^2 \cdot \frac{(s^3 - 3s^2 + 13s - 11)e^{st}}{(s-1-3i)^2 (s-1+3i)^2} \right]^{1}$$

$$= \lim_{x \to 1+3i} \left[\frac{(s^3 - 3s^2 + 13s - 11)e^{st}}{(s-1+3i)^2} \right]^{1} = \dots\dots$$

(a) Fie $F(s)$ not $\mathcal{L}[x(t)](s)$. Aplicând tr Laplace ec

$x'' + x = H(t) - H(t-2\pi)$ obținem:

$\mathcal{L}[x''(t)](s) + \mathcal{L}[x(t)](s) = \mathcal{L}[H(t) - H(t-2\pi)](s)$

$\mathcal{L}[H(t) - H(t-2\pi)](s) = \mathcal{L}[H(t)](s) - \mathcal{L}[H(t-2\pi)](s) =$

$= \frac{1}{s} - \frac{e^{-2\pi s}}{s}$

$\Rightarrow s^2 F(s) - s x(0_+) - x'(0_+) + F(s) = \frac{1 - e^{-2\pi s}}{s}$

$F(s)(s^2+1) - s = \frac{1 - e^{-2\pi s}}{s}$

$F(s) = \frac{1 - e^{-2\pi s} + s^2}{s(s^2+1)} = \frac{s}{s^2+1} + \frac{1}{s^2+1} \cdot \frac{1 - e^{-2\pi s}}{s} \Rightarrow$

$\Rightarrow \mathcal{L}[x(t)](s) = \mathcal{L}[\cos t](s) + \mathcal{L}[\sin t](s) \cdot \mathcal{L}[H(t-2\pi)](s)$

$= \mathcal{L}[\cos t](s) + \mathcal{L}[\sin t * H(t-2\pi)](s) =$

$= [\dots] = \begin{cases} 1 & ; t < 2\pi \\ \cos t & ; t \geq 2\pi \end{cases}.$

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